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COLLAPSIBLE FORM

Abstract

A collapsible concrete form includes a pair of inner panels hingedly coupled to one another and a pair of outer flanges configured to abut a pair of adjacent form sections. A first one of the pair of outer flanges is hingedly coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges is hingedly coupled to a second one of the pair inner panels. The collapsible concrete form further includes a handle protruding from one of the pair of inner panels or a hinge coupling the pair of inner panels. The handle is operable to move the pair of inner panels and the pair of outer flanges between a deployed arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew.

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Background/Summary

[0001] This application claims the benefit of U.S. provisional application entitled “COLLAPSIBLE FORM” filed Mar. 1, 2024, and assigned Ser. No. 63/560,606, the entire disclosure of which is hereby expressly incorporated by reference.

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

[0002] The disclosure generally relates to concrete casting.

Brief Description of Related Technology

[0003] Generally, one or more forms (e.g., steel plywood form systems, aluminum form systems) are used to define a confined space in which concrete is poured to form a concrete wall or casting. After a concrete wall or casting has been poured and has cured to a suitable strength, the form(s) may be stripped (e.g., removed, separated) from the wall or casting. When stripping the forms from the concrete wall or casting, it may be difficult to remove the first form. Specifically, pressure created by the weight of the concrete may cause the forms to become wedged tightly together, making it difficult to remove the first form. Once the first form is stripped or removed, it may be easier to remove subsequent forms from the wall or casting.

SUMMARY OF THE DISCLOSURE

[0004] In accordance with one aspect of the disclosure, a collapsible concrete form includes a pair of inner panels hingedly coupled to one another and a pair of outer flanges configured to abut a pair of adjacent form sections, such that the pair of outer flanges and the pair of inner panels are disposed between the pair of adjacent form sections. A first one of the pair of outer flanges is hingedly coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges is hingedly coupled to a second one of the pair inner panels. The collapsible concrete form further includes a handle protruding from one of the pair of inner panels or a hinge coupling the pair of inner panels. The handle is operable to move the pair of inner panels and the pair of outer flanges between a deployed arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew.

[0005] In accordance with another aspect of the disclosure, a concrete forming system includes a collapsible form having a pair of inner panels hingedly coupled to one another, a pair of outer flanges configured to abut an adjacent form section. A first one of the pair of outer flanges is hingedly coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges is hingedly coupled to a second one of the pair inner panels. The collapsible form further includes a pair of waler bracket mounting holes configured to receive a waler bracket. A first one of the pair of waler bracket mounting holes extends through the first one of the pair of outer flanges and a second one of the pair of waler bracket mounting holes extends through the second one of the pair of outer flanges. The concrete forming system further includes a pair of waler brackets configured to be inserted into the pair of waler bracket mounting holes. The pair of inner panels and the pair of outer flanges are configured to move between a deployed arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew.

[0006] In accordance with another aspect of the disclosure, a collapsible concrete form includes an inner portion including a pair of inner panels pivotably coupled to one another. The inner portion is configured to move between a linear arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner walls are askew. The collapsible form also includes a pair of outer flanges configured to about a pair of adjacent form sections, such that the pair of outer flanges and the inner portion are disposed between the pair of adjacent form sections. A first one of the pair of outer flanges is pivotably coupled to a first one of the pair of inner panels

and a second one of the pair of outer flanges is pivotably coupled to a second one of the pair inner panels.

[0007] In connection with any one of the aforementioned aspects, the devices and systems described herein may alternatively or additionally include any combination of one or more of the following aspects or features. A locking brace may protrude from one of the pair of inner panels. The locking brace may include a pair of projections extending from one of the pair of inner panels and a bearing panel extending between the pair of projections and disposed parallel to one or the pair of inner panels. The collapsible form may include a pair of waler bracket mounting holes configured to receive a waler bracket. A first one of the pair of waler bracket mounting holes may extend through the first one of the pair of outer flanges. A second one of the pair of waler bracket mounting holes may extend through the second one of the pair of outer flanges. The pair of waler bracket mounting holes may be configured to receive a pair of waler brackets. In the deployed arrangement, the pair of waler brackets may be configured to support a rigid member disposed between the pair of waler brackets and the locking brace. The pair of waler brackets and the locking brace are configured to abut the rigid member, preventing the pair of inner panels and the pair of outer flanges from moving from the deployed arrangement. The collapsible form may include a pair of side hinges. A first one of the pair of outer flanges and a first one of the pair of inner panels may collectively form a first one of the pair of side hinges. A second one of the pair of outer flanges and a second one of the pair of inner panels may collectively form a second one of the pair of side hinges. The pair of inner panels may collectively form the hinge coupling the pair of inner panels. In the deployed arrangement, the pair of inner panels are configured to define a planar boundary of confined space in which concrete is poured. A form coupling hole may extend through one of the pair of outer flanges. The form coupling hole may be configured to receive a fastener for coupling a respective outer flange of the pair of outer flanges to an adjacent form section of the pair of adjacent form sections. The bearing panel may be configured to abut a rigid member disposed between the pair of waler brackets and the locking brace. When the rigid member is disposed between the pair of waler brackets and the locking brace, the pair of outer flanges may be configured to abut the rigid member. A handle may protrude from one of the pair of inner panels or a hinge coupling the pair of inner panes. The handle may be operable to move the pair of inner panels and the pair of outer flanges between the deployed arrangement and the collapsed arrangement. A form coupling hole may extend through one of the pair of outer flanges. A collapsible form system may include a form section having a form section panel, a frame coupled to the form section panel and disposed along a perimeter of the form section panel, and a form section coupling hole extending through the frame. A fastener may be configured to be inserted through the form coupling hole and the form section coupling hole, coupling the collapsible form and the form section. When the collapsible form and the form section are coupled, the pair of inner walls and the form section panel are collinear and collectively define planar boundary of a confined space in which concrete is poured. The form section coupling hole may include a pair of form section coupling holes. The fastener may be configured to be inserted through the form coupling hole and one of the pair of form section coupling holes, coupling the collapsible form and the form section. One of the pair of waler brackets may be configured to be inserted through one of the pair of waler bracket mounting holes and an other of the pair of form section coupling holes.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0008] For a more complete understanding of the disclosure, reference should be made to the following detailed description and accompanying drawing figures, in which like reference numerals identify like elements in the figures.

[0009] FIG. 1 illustrates a forming system in accordance with one example of the present disclosure.

[0010] FIG. 2 illustrates a front perspective view of a collapsible form in a deployed arrangement in accordance with one example of the present disclosure.

[0011] FIG. 3 illustrates an exploded view of the collapsible form of FIG. 2 in accordance with one example of the present disclosure.

[0012] FIG. 4 illustrates a back perspective view of the collapsible form of FIG. 2 in accordance with one example of the present disclosure.

[0013] FIG. 5 illustrates a back view of the collapsible form of FIG. 2 in accordance with one example of the present disclosure.

[0014] FIG. 6 illustrates the collapsible form of FIG. 2 in a collapsed arrangement in accordance with one example of the present disclosure.

[0015] FIG. 7 illustrates a collapsible form and a pair of adjacent form sections in accordance with one example of the present disclosure.

[0016] FIG. 8 illustrates a form section in accordance with one example of the present disclosure.

[0017] FIG. 9 illustrates a fastener in accordance with one example of the present disclosure.

[0018] FIG. 10 illustrates a plurality of waler brackets coupled to a collapsible form in accordance with one example of the present disclosure.

[0019] FIG. 11 illustrates a waler bracket in accordance with one example of the present disclosure.

[0020] FIG. 12 illustrates a flow chart for using a collapsible form in accordance with one example of the present disclosure.

[0021] While the disclosed devices and systems are susceptible of embodiments in various forms, there are illustrated in the drawing (and will hereafter be described) specific embodiments of the invention, with the understanding that the disclosure is intended to be illustrative, and is not intended to limit the invention to the specific embodiments described and illustrated herein.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0022] As described above, when a building material, for example, concrete is poured into a confined space defined by a plurality of form sections, the weight of the building material may apply pressure to the form sections. A plurality of fasteners (e.g., pins) may couple the form sections to one another. The pressure of the building material may tightly press the form sections and/or fasteners together. After the building material has at least partially cured, the form sections and fasteners may be removed from the building material (e.g., structure formed by the building material). However, because the form sections and the fasteners have been and/or are tightly pressed together under the pressure (e.g., weight) of the building material, it may be difficult to separate the form sections and/or fasteners from one another and/or from the building material (e.g., a structure formed by the building material). In some examples, separation or removal of a first form section may be most difficult, as there is no space or clearance between the plurality of form sections. After removal of a first form section, an opening or gap may be present between the remaining form sections making removal or separation of subsequent form sections easier.

[0023] A collapsible filler or form is described. The collapsible form may be disposed between two or more form sections. The collapsible form and a plurality of form sections may define a confined space in which a building material is poured and/or at least partially cured. The collapsible form may be configured to move between two or more arrangements. For example, the collapsible form may be configured to move between a deployed arrangement in which the collapsible form defines (e.g., at least a portion of) a boundary of the confined space and a collapsed arrangement. In the collapsed arrangement, the collapsible form may have a smaller or more narrow width than in the deployed arrangement. According to the present disclosure, after pouring and/or curing of a building material (e.g., in the confined space), the collapsible form may transition or move from the deployed arrangement to the collapsed arrangement, such that the collapsible form may be more easily separated from the one or more form sections (e.g., due to the additional space created

between the collapsible form and the form section when the collapsible form is moved from the deployed arrangement to the collapsed arrangement).

[0024] In accordance with some examples, in the deployed arrangement, the collapsible form has a “U” shape including a pair of inner panels, a pair of outer flanges, and three hinges. In accordance with some examples, in the deployed arrangement, the inner panels may form a web of the “U” shape with an outer flange extending from each end of the pair of inner panels. A hinge (e.g., side hinge) may be disposed between each of the outer flanges and an adjacent inner panel. A hinge (e.g., center hinge) may be disposed between the pair of inner panels. The collapsible form may be included in and/or used with a form system (e.g., steel plywood form systems, aluminum form system) to define a confined space in which a concrete wall or casting may be poured. Specifically, the collapsible form may be attached or coupled to one or more steel plywood form sections or aluminum form sections. Specifically, the collapsible form may be disposed between and coupled to two form sections. The collapsible form may be coupled to the one or more form sections by one or more fasteners (e.g., wedge bolts, round pins) extending through the collapsible form and an adjacent form section.

[0025] The pair of inner panels and the pair of side flanges may be hingedly or pivotably coupled to one another, such that the collapsible form may move between a deployed arrangement in which the pair of inner panels are colinear and a collapsed arrangement in which the pair of inner panels are askew. In the deployed arrangement, the pair of inner panels may define a planar boundary of a confined space in which concrete may be poured to form a wall or casting. In some examples, the collapsible form may include a handle operable to move the collapsible form between the deployed and collapsed arrangements. When the collapsed form is in the deployed arrangement a rigid member, for example, a 2”×4” piece of lumber may be placed between a pair of waler brackets coupled to the collapsible form and a locking brace of the collapsible form. The rigid member may abut the waler brackets and the locking brace, preventing the collapsible form (e.g., pair of inner panels) from moving from the deployed arrangement in which the pair of inner panels are colinear while concrete is poured in a confined space defined by the pair of inner panels. The collapsible form may be configured to withstand the hydraulic pressure of concrete poured in a confined space defined by the collapsible form.

[0026] After the concrete has cured, the rigid member, waler brackets, and fasteners coupling the collapsible form to adjacent form sections are removed, and the collapsible form may be moved (e.g., collapsed) from deployed arrangement to the collapsed arrangement. For example, a user may pull on the handle, causing the collapsible form to move from the deployed arrangement to the collapsed arrangement. The collapsible form may have a different (e.g., narrower) shape in the collapsed arrangement than in the deployed arrangement, allowing the collapsible form to be easily removed from a concrete wall or casting. This collapsible form can be used while pouring basements, many different precast castings, above grade walls, or in any casting where pressure from the concrete makes it difficult to remove a first form. The collapsible form can be used with steel plywood forms, aluminum forms, or any other type of form.

[0027] Referring to FIG. 1, a forming system **100** is illustrated in accordance with one example of the preset disclosure. As shown in FIG. 1, the form system **100** may include a collapsible form **110**, two or more form sections **120**, two or more waler brackets **130**, and at least one rigid member **140**. In accordance with some examples, the forming system **100** may be a concrete forming system. According to other examples, the forming system **100** may be used with another type of building material, for example, plaster, asphalt, or the like. According to the present disclosure, the collapsible form **110** may be removably coupled to two or more form sections **120**. For example, as shown in FIG. 1, the collapsible form **110** may be directly coupled to and disposed between a pair of adjacent form sections **120**.

[0028] Referring to FIG. 2, a back perspective view of the collapsible form **110** is illustrated in accordance with one example of the present disclosure. The collapsible form **110** may include two

or more inner panels **210** and two outer flanges **220**. According to some examples, as shown in FIG. **1**, the collapsible form **110** may include a pair of (e.g., two) inner panels **210**. However, the present disclosure is not limited thereto and in other examples, another number of inner panels **210** may be included. For example, three, four, or more than four inner panels **210** may be included. [0029] As noted above, the collapsible form **110** may be configured to move between a deployed arrangement in which the inner panels **210** are collinear and a collapsed arrangement in which the inner panels **210** are askew (e.g., disposed at a non-zero angle with respect to one another). As shown in FIG. **2**, the collapsible form **110** is disposed in a deployed arrangement.

[0030] In the deployed arrangement, the inner panels **210** may define (e.g., at least a portion of) a (e.g., planar) boundary of a confined space in which a building material, for example, concrete, asphalt, plaster, or the like, may be poured. Additionally, the building material may be at least partially cured within the confined space, a boundary of which is at least partially defined by the inner panels **210** when the collapsible form **110**. For example, the building material may be cured within the confined space at least until the building material is self-supporting. The confined space may correspond, for example, in size and/or shape, to a structure to be formed by the building material.

[0031] In the collapsed arrangement, a width of the collapsible form **110**, for example, a distance between the outer flanges **220** of the collapsible form **110** may be smaller than a width of the of the collapsible form **110** in the deployed arrangement. Accordingly, as noted above, after a building material has been poured into and at least partially cured within the confined space, collapsible form **110** may be moved (e.g., collapsed) from the deployed arrangement to the collapsed arrangement before or as the collapsible form **110** is separated from and/or removed from between the pair of adjacent form sections. As described above, moving the collapsible form from the deployed arrangement to the collapsed arrangement may facilitate or improve the ease with which the collapsible form **110** may be separated from and/or removed from between the pair of adjacent form sections.

[0032] Similarly, in some examples, the collapsible form **110** may be inserted between the pair of adjacent form sections, for example, before pouring of the building material, in the collapsed arrangement and moved (e.g., expanded) to the deployed arrangement while already disposed between the pair of adjacent form sections.

[0033] Still referring to FIG. **2**, the inner panels **210** may be movably coupled to one another. For example, the inner panels **210** may be hingedly or pivotably coupled to one another, such that the inner panels **210** are free to move relative to one another. As described hereinafter in greater detail, the inner panels **210** may be movably coupled to one another so as to be collinear when the collapsible form **110** is in a deployed arrangement and askew when the collapsible form **110** is in a collapsed arrangement. One or more center hinge(s) **230** may rotatably couple the inner panels **210**.

[0034] FIG. **3** illustrates an exploded view of the collapsible form **110** of FIG. **2**. Referring to FIGS. **2** and **3**, the center hinge **230** may include a barrel **231** and a pin **232**. The barrel **231** may be comprised of a plurality of barrel segments **233**. The barrel **231** may be comprised of one or more barrel segments **233** fixedly attached to a first one of the pair of inner panels **210** and one or more barrel segments **233** fixedly attached to a second one of the pair of inner panels **210**. According to some examples, the one or more barrel segments **233** fixedly attached to the first one of the pair of inner panels **210** and the one or more barrel segments **233** fixedly attached to the second one of the pair of inner panels **210** may be integrally formed with their respective inner panel **210**. According to other examples, one or more barrel segments **233** may be fixedly attached to a respective one of the pair of inner panels **210** using one or any combination of: one or more fasteners (e.g., nuts, bolts, rivets, screws, or the like), welding, adhesive bonding, or the like.

[0035] The one or more barrel segments **233** fixedly attached to the first one of the pair of inner panels **210** and the one or more barrel segments **233** fixedly attached to the second one of the pair of inner panels **210** may be moved or positioned such that a hole or channel extending through each

of the barrel segments **233** is aligned. The pin **232** may be inserted or disposed within the channel of the plurality of barrel segments **233**, movably coupling the pair inner panels **210** to one another. For example, the pair of inner panels **210** may be coupled such that they may rotate or pivot relative to one another about the pin **232**.

[0036] FIG. **4** illustrates a front perspective view of the collapsible form **110** of FIG. **2** in a deployed state. The inner panels **210** may be configured to define (e.g., at least a portion of) a boundary of a confined space in which concrete is poured. For example, when the collapsible form **110** is in the deployed state, as shown in FIGS. **2** and **4**, the inner panels **210** may be configured to define (e.g., at least a portion of) a boundary of a confined space in which concrete is poured. A front surface or face **211** of the inner panels **210** may be configured to define (e.g., at least a portion of) a boundary of a confined space in which concrete is poured. According to some examples, the face **211** of the inner panels **210** may be planar, such that the portion of a boundary of a confined space in which concrete may be poured defined by each of the inner panels **210** is planar. According to the present disclosure, the inner panels **210** of the collapsible form **110** may be collinear when the collapsible form **110** is in the deployed arrangement, such that the inner panels **210** collectively define a planar boundary of the confined space.

[0037] Returning to FIG. **2**, the collapsible form **110** may further include one or more locking braces **260**. As shown in FIG. **1**, when the collapsible form **110** is disposed between a pair of adjacent form sections **120**, two or more waler brackets **130** may be coupled to the collapsible form **110**. The waler brackets **130** may be (e.g., removably) coupled to the collapsible form **110** in pairs. Each pair of waler brackets **130** may be configured to support a rigid member **140** disposed between the pair of waler brackets **130** and one or more locking braces **260** extending from the collapsible form **110**. The one or more locking braces **260** may be configured to abut a rigid member **140** disposed between a pair of waler brackets **130** and the collapsible form **110**. The one or more locking braces **260** may exert a force or bear on the rigid member **140** (e.g., supported by the pair of waler brackets **130** and disposed between the pair of waler brackets **130** and the collapsible form **110**), preventing the collapsible form **110** from moving from the deployed arrangement (e.g., to the collapsed arrangement).

[0038] According to the present disclosure, when a building material is poured in a confined space having a boundary at least partially defined by the inner panels **210** of the collapsible form **110**, pressure created by a weight of the building material may exert an (e.g., outward) force on the inner panels **210**, biasing the collapsible form **110** to move from the deployed arrangement, in which the inner panels **210** are collinear, to the collapsed arrangement, in which the inner panels **210** are askew. However, when the rigid member **140** is disposed between a pair of waler brackets **130** and the one or more locking braces **260**, the rigid member **140** may exert a force on the one or more locking braces **260** that is equal to and opposite the outward force exerted (e.g., due to the pressure of the building material in the confined space) on the inner panels **210**, preventing the collapsible form **110** from moving from the deployed arrangement.

[0039] Returning to FIG. **2**, each of the one or more locking braces **260** may extend from a respective one of the inner panels **210**. Each of the one or more locking braces **260** may include at least one projection **261** extending from one of the pair of inner panels **210** and a bearing surface or bearing panel **262** configured to engage (e.g., abut, contact) a rigid member disposed between a pair of waler brackets and the collapsible form **110**. According to some examples, as shown in FIG. **2**, each locking brace **260** may include two projections **261** extending from one of the inner panels **210**. According to some examples, as shown in FIG. **2**, the bearing panel **262** may extend between the pair of projections **261**. However, the present disclosure is not limited thereto, and in other examples, the bearing panel **262** may be coupled to a single projection **261** included in the locking brace **260**. According to some examples, the bearing panel **262** may be disposed parallel to an inner panel **210** to from which the locking brace **260** extends.

[0040] Still referring to FIG. **2**, the collapsible form **110** may include a pair of outer flanges **220**.

The pair of outer flanges **220** may be configured to engage a pair of adjacent form sections, for example, the adjacent form sections **120** illustrated in FIG. 1, when the collapsible form **110** is in the deployed arrangement. Each of the outer flanges **220** may engage a respective one of the pair of adjacent form sections, such that the collapsible form **110** is disposed between the pair of adjacent form sections. For example, each of the pair of outer flanges **220** may be configured to abut or contact a respective adjacent form section. In accordance with some examples, each outer flange **220** may include a contact surface **221** configured to interface with engage (e.g., abut, contact) a form section adjacent to the respective outer flange **220**.

[0041] As described above, the collapsible form **110** may be configured to move between the deployed arrangement (as illustrated in FIG. 2) and a collapsed arrangement. According to the present disclosure, each of the pair of outer flanges **220** may be movably coupled to a respective one of the inner panels **210**, facilitating movement of the collapsible form **110** between the deployed and collapsed arrangements. For example, a first one of the pair of outer flanges **220** may be movably coupled to a first one of the inner panels **210** and a second one of the pair of outer flanges **220** may be movably coupled to a second one of the inner panels **210**. For example, the outer flanges **220** may be hingedly or pivotably coupled to a respective one of the inner panels **210**, such that the outer flange **220** is free to move relative to a respective one of the inner panels **210**.

[0042] According to some examples, when the collapsible form **110** is in the deployed arrangement, as illustrated in FIG. 2, the pair of outer flanges **220** may be disposed parallel to one another.

[0043] Referring to FIGS. 2 and 3, according to some examples, the collapsible form **110** may include a pair of side hinges **240** movably coupling the outer flanges **220** to their respective inner panels **210**.

[0044] Each side hinge **240** may include a barrel **241** and a pin **242**. The barrel **241** may be comprised of a plurality of barrel segments **243**. The barrel **241** may be comprised of one or more barrel segments **243** fixedly attached to a respective one of the outer flanges **220** and one or more barrel segments **243** fixedly attached to a respective one of the inner panels **210**. According to some examples, the one or more barrel segments **243** fixedly attached to the outer flange **220** and the one or more barrel segments **243** fixedly attached to the inner panel **210** may be integrally formed with the outer flange **220** and/or inner panel **210**, respectively. According to other examples, the one or more barrel segments **243** may be fixedly attached to the outer flange **220** and/or the one or more barrel segments **243** may be fixedly attached to the inner panel **210** using one or any combination of: one or more fasteners (e.g., nuts, bolts, rivets, screws, or the like), welding, adhesive bonding, or the like.

[0045] The one or more barrel segments **243** fixedly attached to the outer flange **220** and the one or more barrel segments fixedly attached to the inner panel **210** may be aligned. For example, the one or more barrel segments **243** fixedly attached to the outer flange **220** and the one or more barrel segments fixedly attached to the inner panel **210** may be aligned such that a hole or channel extending through each of the barrel segments **243** is aligned. The pin **242** may be inserted or disposed within the channel of the plurality of barrel segments **243**, movably coupling the outer flange **220** and the inner panel **210**. For example, the outer flange **220** and the inner panel **210** may be coupled such that they may rotate or pivot relative to one another about the pin **242**.

[0046] FIG. 5 illustrates a back view of the collapsible form **110** of FIG. 1 in accordance with one example of the present disclosure.

[0047] Referring to FIG. 6, the collapsible form **110** of FIG. 2 is illustrated in a collapsed state in accordance with one example of the present disclosure. As noted above, the collapsible form **110** may be configured to move between a deployed arrangement (e.g., illustrated in FIGS. 2, 4, and 5) and a collapsed arrangement (e.g., illustrated in FIG. 6). As described above, the inner panels **210** may be movably coupled to one another and the outer flanges **220** may be movably coupled to a respective one of the inner panels **210**. The movable connections or couplings between the inner panels **210** and the outer flanges **220** and a respective one of the inner panels **210** described above

may facilitate or allow movement of the collapsible form **110** between the deployed arrangement (e.g., as illustrated in FIGS. **2**, **4**, and **5**) and the collapsed arrangement (e.g., as illustrated in FIG. **6**).

[0048] Referring to FIG. **2**, in some examples, the collapsible form **110** may further include a handle **250**. The handle **250** may be operable to move the collapsible form **110** between the deployed and collapsed arrangements. For example, the handle **250** may be grabbed or held by a person who may exert a force (e.g., push, pull) on the collapsible form **110**, via the handle **250**, moving the collapsible form **110** between the deployed and collapsed arrangements. According to some examples, a person may grab and hold one of an inner panel **210** or an outer flange **220** (e.g., with a second hand) while a force is exerted on the collapsible form **110** via the handle **250**, such that, the collapsible form **110** is moved between the deployed and collapsed arrangements. According to other examples, one or both of the outer flanges **220** may be coupled to an adjacent form section while a force is exerted on the collapsible form **110** via the handle **250**, such that, the collapsible form **110** is moved between the deployed and collapsed arrangements.

[0049] According to some examples, as shown in FIG. **2**, the handle **250** may be coupled to the center hinge **230**. For example, the handle **250** may be coupled to a barrel segment **233** coupled to one of the pair of inner panels **210**. According to some examples, the handle **250** may be coupled to one of the inner panels **210**, for example, to a back surface of one of the inner panels **210**.

[0050] As shown in FIG. **6**, in the collapsed arrangement, the inner panels **210** of the collapsible form **110** may be askew or disposed at an angle (e.g., other than zero) with respect to one another. According to the present disclosure, a width of the collapsible form **110** may change as the collapsible form **110** moves between the deployed and collapsed arrangements. The width of the collapsible form **110** may be defined as a distance between the pair of outer flanges **220**. For example, the width of the collapsible form **110** may be a distance between the contact surfaces **221** of the pair of outer flanges **220**. According to the present disclosure, because the inner panels **210** are askew in the collapsed arrangement, a width of the collapsible form **110** in the collapsed arrangement may be smaller than a width of the collapsible form **110** in the deployed arrangement.

[0051] Referring to FIG. **7**, the collapsible form **110** may be coupled to one or more form sections **120**. For example, as described above and illustrated in FIG. **7**, the collapsible form **110** may be disposed between a pair of adjacent form sections **120**. The collapsible form **110** and the one or more form sections **120** may collectively define the confined space in which a building material may be poured and/or (e.g., at least partially) cured. In some examples, the collapsible form **110** may be (e.g., directly) coupled to a pair of adjacent form sections **120**. As described above, each of the outer flanges **220** may be configured to engage or interface with an (e.g., directly) adjacent form section **120**, such that the collapsible form **110** is disposed between a pair of (e.g., directly) adjacent form sections.

[0052] According to the present disclosure, the collapsible form **110** may be configured to move between the collapsed arrangement (e.g., as shown in FIG. **6**) and the deployed arrangement (e.g., as shown in FIGS. **2** and **7**), such that the collapsible form **110** may be inserted between and/or removed from between a pair of (e.g., directly) adjacent form sections **120** in the collapsed arrangement and define (e.g., at least a portion of) a boundary of a confined space in the deployed arrangement.

[0053] Referring to FIG. **7**, according to the present disclosure, a distance between a pair of adjacent form sections **120**, between which the collapsible form **110** is inserted or disposed may be equal to or less than a width of the collapsible form **110** when the collapsible form **110** is in the deployed arrangement. Accordingly, the collapsible form **110** may be inserted between the pair of adjacent form sections, in the collapsed arrangement and as the collapsible form **110** is moved to the deployed arrangement, the pair of outer flanges **220** may be brought into contact with respective ones of the pair of adjacent form sections **120**. In accordance with some examples, a distance between the pair of adjacent form sections **120**, between which the collapsible form **110** is

inserted may be smaller than a width of the collapsible form **110** in the deployed arrangement, such that when the collapsible form **110** (e.g., in the deployed arrangement) is disposed between the pair of adjacent form sections **120**, the collapsible form **110** and the pair of adjacent form sections **120** bear or exert a force on one another, for example, such that relative positions of the collapsible form **110** and the form sections **120** are maintained.

[0054] Referring to FIG. 8, a form section **120** is illustrated in accordance with one example of the present disclosure. As shown in FIG. 8, the form section **120** may include a frame **810** and a form section panel **820**. The form section panel **820** may be configured to define (e.g., at least a portion of) a boundary of a confined space in which a building material is poured and/or (e.g., at least partially) cured. According to some examples, the form section panel **820** may have a planar face configured to define a boundary of the confined space. According to some examples, the form section panel **820** may be composed of wood. For example, the form section panel **820** may be composed of plywood (e.g., one or more layers of wood glued to one another). According to other examples, the form section panel **820** may be composed of a metal or metal alloy. For example, the form section panel may be composed of aluminum, an aluminum alloy, steel, or another metal.

[0055] The frame **810** may be disposed behind the form section panel **820**, for example, outside of the confined space. The frame **810** may be coupled to the form section panel **820**. The frame **810** may be configured to support the form section panel **820**. According to some examples, as shown in FIG. 8, the frame **810** may be disposed along a perimeter or periphery of the form section panel **820**. The frame **810** may be comprised of a plurality of linear members or beams **811** coupled to one another. According to some examples, as shown in FIG. 8, a beam **811** of the frame **810** may extend along each side of the form section panel **820**. According to some examples, the frame **810** may further include one or more (e.g., horizontal, vertical, diagonal) beams **811** extending between two of the beams **811** extending along respective sides of the form section panel **820**. According to some examples, the frame **810** (e.g., the beams **811** included in the frame **810**) may be composed of a metal or metal alloy. For example, the frame **810** (e.g., beam **811**) may be composed of aluminum, an aluminum alloy, steel, or another metal. According to other examples, the frame **810** (e.g., beams **811**) may be composed of wood.

[0056] The size and shape of the form section **120** (e.g., the frame **810** and form section panel **820**) may vary. According to some examples, the form section **120** may have a rectangular (e.g., square) shape. According to other examples, the form section **120** may have another (e.g., polygonal) shape. A height of the form section **120** may vary. For example, the form section **120** may be 3 feet, 4 feet, 5 feet, 6 feet, 7 feet, 8 feet, 9 feet, or more than 9 feet tall.

[0057] In some examples, the collapsible form **110** may be (e.g., removably) coupled to one or more form sections **120**. For example, each outer flange **220** may be (e.g., removably) coupled to a respective one of the pair of adjacent form sections **120**.

[0058] An outer flange **220** of the collapsible form **110** and a form section **120** (e.g., directly) adjacent to the outer flange **220** may be coupled to one another. For example, as shown in FIGS. 2, 3, and 7, each outer flange **220** may include one or more coupling holes **223** extending therethrough. Additionally, as shown in FIG. 8, the form section **120**, for example, the frame **810**, may include one or more form section coupling holes **813** extending therethrough. As shown in FIG. 7, when the collapsible form **110** is disposed between a pair of adjacent form sections **120** and in the deployed arrangement, one or more form coupling holes **223** extending through an outer flange **220** may align with the one or more form section coupling holes **813** extending through the adjacent form section **120** and a fastener may be inserted through the aligned holes, coupling the collapsible form **110** and the form section **120**.

[0059] For example, the fastener **900**, illustrated in FIG. 9, may be inserted through the aligned holes, coupling the collapsible filler **110** and the form section **120**. As shown in FIG. 9, the fastener **900** may include a body **910** and a head **920** disposed at an end of the body **910**. The fastener **900** may be inserted through an aligned form coupling hole **223** and form section coupling hole **813**, for

example, until the head **920** of the fastener **900** abuts one of the outer flange **220** of the collapsible form **110** or the frame **810** of the form section **120**. According to other examples, another fastener, such as, a pin, bolt, rivet, or the like, may be inserted through the aligned holes, coupling the collapsible form **110** and the form section **120**.

[0060] FIG. **10** illustrates a plurality of waler brackets **130** coupled to a collapsible form **110** in accordance with one example of the present disclosure. In some examples, the plurality of waler brackets **130** may be coupled to the collapsible form **110** in pairs. As shown in FIG. **1**, each pair of waler brackets **130** may be configured to support or hold a rigid member **140** disposed between the pair of waler brackets **130** and the collapsible form **110**. According to the present disclosure, one (e.g., a first one) of the pair of waler brackets **130** may be (e.g., directly) coupled to one (e.g., a first one) of the pair of outer flanges **220** and an other (e.g., a second one) of the pair of waler brackets **130** may be (e.g., directly) coupled to an other one (e.g., second one) of the pair of outer flanges **220**. As described above, when the rigid member **140** is disposed between the collapsible form **110** and the pair of waler brackets **130**, the rigid member **140** may prevent the collapsible form **110** from moving from the deployed arrangement.

[0061] Referring to FIG. **11**, a waler bracket **130** is illustrated in accordance with one example of the present disclosure. As shown in FIG. **11**, the waler bracket **130** may include a waler bracket body or a body **1110** and a waler bracket pin or a pin **1120**. The body **1110** may be (e.g., directly, removably) coupled to the collapsible form, for example, collapsible form **110** illustrated in FIG. **2**. The pin **1120** may be (e.g., directly, removably) coupled to the body **1110**.

[0062] As shown in FIG. **11**, the body **1110** may include a first or upper shelf **1111** and a second or lower shelf **1112**. In accordance with some example, the upper shelf **1111** may define an upper surface of the waler bracket **130** configured to (e.g., directly) support a rigid member, for example, the rigid member **140** illustrated in FIG. **1**. As shown in FIG. **11**, the upper shelf **1111** may include a tab **1113**. The tab **1113** may be inserted through a hole included in a collapsible form (e.g., collapsible form **110**) and/or a form section (e.g., form section **120**) removably coupling the waler bracket **130** to the collapsible form and/or the form section.

[0063] Returning to FIG. **2**, the collapsible form **110** may include one or more waler bracket mounting holes **225**. For example, as shown in FIG. **2**, one or more waler bracket mounting holes **225** may extend through each of the outer flanges **220**. The one or more waler bracket mounting holes **225** may be configured to receive a tab of the waler bracket, for example, tab **1113** illustrated in FIG. **11**, such that the waler bracket may be removably coupled to the collapsible form **110**.

[0064] In accordance with some examples, the waler bracket mounting holes **225** and the form coupling holes **223** may have the same size and/or shape, such that, the waler bracket mounting holes **225** may be configured to receive a fastener, for example, fastener **900** illustrated in FIG. **9**, for coupling the collapsible form **110** to an adjacent form section and/or the form coupling holes **223** may be configured to receive a tab for coupling a waler bracket to the collapsible form **110**. In accordance, with other examples, the waler bracket mounting holes **225** and the form coupling holes **223** may have different size and/or shapes, such that the waler bracket mounting holes **225** are not capable of receiving a fastener, for example, fastener **900** illustrated in FIG. **9**, for coupling the collapsible form to an adjacent form section and the form coupling holes **223** are not cable of receive a tab for coupling a waler bracket to the collapsible form **110**.

[0065] Returning to FIG. **11**, the tab **1113** may be configured to be removably inserted into a waler bracket mounting hole directly coupling the body **1110** to the collapsible form. In accordance with some examples, the waler bracket **130** may be (e.g., removably, directly) coupled to both a collapsible form and a form section (e.g., adjacent to the collapsible form). For example, a tab **1113** of the waler bracket **130** may extend through both a waler bracket mounting hole (e.g., **225** illustrated in FIG. **2**) of the collapsible form and a form section coupling hole (e.g., **813** illustrated in FIG. **8**, which are aligned).

[0066] The lower shelf **1112** may include a wall **1114** extending (e.g., downward) from the lower

shelf **1112**. The wall **1114** may be configured to abut an outer flange of the collapsible form when the waler bracket **130** is coupled to the collapsible form. For example, as shown in FIG. **10**, when the tab **1113** of the waler bracket **130** extends through a waler bracket mounting hole **225**, the wall **1114** of the waler bracket **130** may contact an outer flange **220** of the collapsible form **110**. In some examples, the wall **1114** may also contact the frame of a form section (e.g., directly) adjacent to the collapsible form **110**.

[0067] Returning to FIG. **11**, the pin **1120** may be removably (e.g., directly) coupled to the body **1110** of the waler bracket **130**. As shown in FIG. **11**, in some examples, the pin **1120** may have a substantially “U” shape including a web **1121** and a pair of legs **1122** extending (e.g., in the same direction) from opposite ends of the web **1121**. In accordance with some examples of the present disclosure, when the pin **1120** is coupled to the body **1110**, the web **1121** of the pin **1120** may be configured to abut a rigid member, for example, rigid member **140** illustrated in FIG. **1**, disposed between the waler bracket **130** (e.g., the pin **1120**) and the collapsible form (e.g., a locking brace and/or outer flanges).

[0068] In accordance with some examples, the pin **1120** may be inserted through a slot or an upper slot **1115** extending through the upper shelf **1111** and/or a slot or lower slot **1116** extending through the lower shelf **1112**. In accordance with some examples, the upper slot **1115** and/or lower slot **1116** may have a shape corresponding to the pin **1120**. For example, as shown in FIG. **11**, the upper slot **1115** may include a web portion **1118** configured to receive the web **1121** of the pin **1120** and two leg portions **1119** configured to receive respective legs **1122** of the pin **1120**.

[0069] Returning to FIG. **1**, the forming system **100** further includes a rigid member **140**. The rigid member **140** may be configured to be provided (e.g., placed, positioned) between a pair of waler brackets **130** and the collapsible form **110**. The rigid member **140** may be configured to abut the waler bracket pins, for example, waler bracket pins **1120** illustrated in FIG. **11**, and the one or more locking braces, for example, locking braces **260** illustrated in FIG. **2**, when the rigid member **140** is disposed between the pair of waler brackets **130** and the collapsible form **110**. According to some examples, as shown in FIG. **1**, the rigid member **140** may contact an end of each of the pair of outer flanges of the collapsible form **110** when the rigid member is disposed between the pair of waler brackets **130** and the collapsible form **110**. The rigid member **140** (e.g., when disposed between the pair of waler brackets **130** and the collapsible form) may be configured to prevent the collapsible form **110** from moving from the deployed arrangement to the collapsed arrangement. The rigid member **140** may be rigid, such that, the rigid member **140** does not deform in response to a force applied to the rigid member **140** by the waler brackets **130** and/or collapsible form **110**.

[0070] The size and/or shape of the rigid member **140** may vary. In some examples, the size and/or shape of the rigid member **140** may vary in accordance with a size and/or shape of the pair of waler brackets **130** coupled to the collapsible form **110**. For example, the rigid member **140** may have a size and/or shape (e.g., about at least one axis) corresponding to a distance between the pair of waler bracket pins and the collapsible form **110** (e.g., the one or more locking braces and/or a back end of the outer flanges). According to some examples, the rigid member **140** may be composed of wood. For example, the rigid member **140** may be a piece of lumber (e.g., a 2"×4" piece of lumber, 1.5"×3.5" piece of lumber, any other size piece of lumber). According to other examples, the rigid member **140** may be composed of another material, for example, a metal or metal alloy, a plastic or plastics, or the like.

[0071] Referring to FIG. **12**, a flowchart **1200** for using a collapsible form, for example, collapsible form **110** illustrated in FIG. **2**, and/or a forming system, for example, forming system **100** illustrated in FIG. **1**, is illustrated in accordance with one example of the present disclosure. Additional, different, or fewer acts may be provided.

[0072] At **1201**, a collapsible form may be positioned or placed between a pair of adjacent form sections. For example, a user (e.g., person) may place the collapsible form between a pair of adjacent form sections. In accordance with some examples, the collapsible form may be in the

collapsed arrangement when the collapsible form is placed between the pair of adjacent form sections. Accordingly, a space (e.g., opening, gap) may be disposed between the collapsible form and one or both of the form sections when the collapsible form is in the collapsed arrangement. [0073] At **1203**, the collapsible form may transition or move from the collapsed arrangement to the deployed arrangement. For example, a user may move the collapsible form from the collapsed arrangement to the deployed arrangement using a handle of the collapsible form. In accordance with some examples, as the collapsible form is moved to the deployed arrangement, the collapsible form may engage (e.g., abut, contact) the pair of adjacent form sections. For example, the outer flanges of the collapsible form may be brought into contact with a respective one of the pair of adjacent form sections.

[0074] At **1205**, one or more fasteners, for example, fastener **900** illustrated in FIG. **9**, may be inserted through aligned holes of the collapsible form and a respective one of the pair of adjacent form sections. For example, the fastener may be inserted through a hole, for example, a form coupling hole (e.g., **223** as illustrated in FIG. **2**) extending through an outer flange of the collapsible form and a form section coupling hole (e.g., **813** as illustrated in FIG. **8**), coupling the collapsible form to a respective one of the pair of adjacent form sections.

[0075] At **1207**, two or more waler brackets, for example, waler brackets **130** illustrated in FIG. **11**, may be coupled to the collapsible form and/or the pair of adjacent form sections. For example, a tab of the waler bracket may be inserted through a waler bracket mounting hole extending through the collapsible form. In some examples, the tab may be inserted through a waler bracket mounting hole of the collapsible form and a form section coupling hole extending through the frame of a form section (e.g., directly) adjacent to the collapsible form, such that the waler bracket is (e.g., directly, removably) coupled to both the collapsible form and form section.

[0076] Additionally, the pins, for example, pins **1120** illustrated in FIG. **11**, may be inserted into the body, for example, body **1110** illustrated in FIG. **11**, of the waler bracket at **1207**.

[0077] At **1209**, a rigid member, for example, rigid member **140** illustrated in FIG. **1**, may be provided between a pair of waler brackets and the collapsible form. For example, the rigid member may be provided (e.g., placed, positioned) between the respective pins of a pair of waler brackets and the collapsible form (e.g., locking braces and/or outer flanges of the collapsible form).

[0078] At **1211**, the building material may be poured into and (e.g., at least partially) cured within the confined space defined by the collapsible form and the plurality of form sections. As described above, according to some examples, concrete may be poured and at least partially cured within a confined space defined by the collapsible form (disposed in the deployed arrangement) and one or more form sections. In accordance with other examples, plaster, asphalt, or another building material may be poured and at least partially cured within the confined space.

[0079] At **1213**, the pins of the waler bracket may be removed from the body of the waler bracket. According to the present disclosure, when the building material (e.g., concrete) is poured and (e.g., at least partially) cured within the confined space, a weight or pressure of the building material may exert an outward force on the collapsible form, for example, the inner panels of the collapsible form. In some examples, the weight or pressure of the building material may cause the collapsible form, rigid member, and waler bracket to be wedged tightly together. In accordance with some examples, the pins, for example, pins **1120** illustrated in FIG. **11**, may first be separated or decoupled from the pair of waler brackets, such that a rigid member is free to move away from the collapsible form. In some examples, a force may be applied to (e.g., a bottom of) the pins, such that the pins may be pushed upward and out of slots (e.g., upper and/or lower slots) included in the body of the waler bracket. For example, a hammer or another tool may be used to strike a bottom of the pins, pushing the pins upward and out of their respective slots in the body of the waler bracket.

[0080] At **1215**, the rigid member may be removed or separated from the waler brackets and/or the collapsible form. According to some examples, as described above, the pins of the waler bracket

may be removed before the rigid member is separated from the waler brackets and/or the collapsible form, in these examples, the rigid member may no longer be disposed between the collapsible form and the pins of the waler bracket such that the rigid member is free to simply be moved away from the collapsible form and the body of the waler brackets. In accordance with other examples, the waler bracket pins may still be coupled to the waler bracket body when the rigid member is separated from the collapsible form and the waler brackets. In these examples, the rigid member may be moved upward so as to be separated from the waler brackets and/or collapsible form.

[0081] At **1217**, the fasteners may be removed from aligned holes of the collapsible form and the pair of adjacent form sections. After removal of the fasteners the collapsible form and adjacent form sections may be free move relative to (e.g., be separated from) one another.

[0082] At **1219**, the collapsible form may move or transition from the deployed arrangement to the collapsed arrangement. For example, a user may move the collapsible form from the deployed arrangement to the collapsed arrangement. According to the present disclosure, movement of the collapsible form from the deployed arrangement to the collapsed arrangement may facilitate removal or separation of the collapsible form from the pair of adjacent form sections and/or a structure formed by the building material. In accordance with some examples, the collapsible form may be moved to the collapsed arrangement before being moved from between the pair of adjacent form sections. In accordance with other examples, the collapsible form may be moved to the collapsed arrangement as it is being removed from between the pair of adjacent form sections. In accordance with one example of the present disclosure, the collapsible form may be provided as a 12" wide "U" shaped panel. In other examples, the collapsible form may be provided as a 4" wide "U" shaped panel. In accordance with some examples of the present disclosure, the collapsible form may be 3', 4', 5', 6', 8', or 9' tall.

[0083] According to some examples of the present disclosure, the collapsible form may be comprised of a metal or metal alloy. In some examples, the collapsible form may be comprised of steel. In other examples, the collapsible form may be comprised of aluminum or an aluminum alloy. In some examples, the collapsible form and the form sections (e.g., form system) the collapsible form is intended to be used with may be comprised of the same material. For examples, a collapsible form configured to be used with steel plywood form sections may be comprised of steel. Similarly, a collapsible form configured to be used with aluminum form sections may be comprised of aluminum or aluminum alloy.

[0084] According to the present disclosure, the collapsible form may be compatible with (e.g., configured to be coupled to) any currently existing or later developed steel plywood forms or forming systems. For example, the collapsible form may be compatible with EMI steel plywood forms, Symons steel plywood forms, SureBuilt Handset steel plywood forms, and the like.

[0085] According to the present disclosure, the collapsible form may be compatible with (e.g., configured to be coupled to) any currently existing or later developed aluminum forming systems. For example, the collapsible form may be compatible with EMI aluminum forms, Western Forms aluminum forms, Precise Forms aluminum forms, and the like.

[0086] The number of waler brackets coupled to the collapsible form (and thus a number of waler bracket mounting holes) may vary. An even number of waler brackets may be coupled to the collapsible form and an equal number of waler brackets may be coupled to each of the pair of outer flanges. The number of rigid members coupled to the collapsible form may correspond to the number of waler brackets coupled to each of the outer flanges. The number of waler brackets coupled to each of the pair outer flanges may vary in accordance with a height of the collapsible form; taller collapsible forms may include more waler brackets.

[0087] Two or more collapsible forms may used for a single concrete pour or casting. Specifically, two or more collapsible forms may define a boundary of the same confined space in which concrete is poured. In some examples, the two or more collapsible forms may define opposite boundaries of

a confined space in which concrete is poured. In some examples, the two or more collapsible forms may define a boundary of different walls included in a confined space in which concrete is poured. [0088] In accordance with some examples of the present disclosure, each of the pair of outer flanges may include one or more waler bracket mounting holes configured to receive a waler bracket and one or more form coupling holes configured to receive a fastener for coupling the collapsible form to an adjacent form section. In some examples, the waler bracket mounting holes and form coupling holes may have the same shape and/or size such that the waler bracket mounting holes and form coupling holes both be configured to receive a waler bracket and a fastener. [0089] While the present invention has been described with reference to specific examples, which are intended to be illustrative only and not to be limiting of the invention, it will be apparent to those of ordinary skill in the art that changes, additions and/or deletions may be made to the disclosed embodiments without departing from the spirit and scope of the invention. [0090] The foregoing description is given for clarity of understanding only, and no unnecessary limitations should be understood therefrom, as modifications within the scope of the invention may be apparent to those having ordinary skill in the art.

Claims

1. A collapsible concrete form comprising: a pair of inner panels hingedly coupled to one another; a pair of outer flanges configured to abut a pair of adjacent form sections, such that the pair of outer flanges and the pair of inner panels are disposed between the pair of adjacent form sections, a first one of the pair of outer flanges hingedly coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges hingedly coupled to a second one of the pair inner panels; and a handle protruding from one of the pair of inner panels or a hinge coupling the pair of inner panels, the handle operable to move the pair of inner panels and the pair of outer flanges between a deployed arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew.
2. The collapsible concrete form of claim 1, further comprising: a locking brace protruding from one of the pair of inner panels.
3. The collapsible concrete form of claim 2, wherein the locking brace includes: a pair of projections extending from one of the pair of inner panels; and a bearing panel extending between the pair of projections and disposed parallel to one or the pair of inner panels.
4. The collapsible concrete form of claim 2, further comprising: a pair of waler bracket mounting holes configured to receive a waler bracket, a first one of the pair of waler bracket mounting holes extending through the first one of the pair of outer flanges and a second one of the pair of waler bracket mounting holes extending through the second one of the pair of outer flanges, wherein the pair of waler bracket mounting holes are configured to receive a pair of waler brackets and in the deployed arrangement, the pair of waler brackets are configured to support a rigid member disposed between the pair of waler brackets and the locking brace and the pair of waler brackets and the locking brace are configured to abut the rigid member, preventing the pair of inner panels and the pair of outer flanges from moving from the deployed arrangement.
5. The collapsible concrete form of claim 1, further comprising: a pair of side hinges; and wherein a first one of the pair of outer flanges and a first one of the pair of inner panels collectively form a first one of the pair of side hinges, wherein a second one of the pair of outer flanges and a second one of the pair of inner panels collectively form a second one of the pair of side hinges, and wherein the pair of inner panels collectively form the hinge coupling the pair of inner panels.
6. The collapsible concrete form of claim 1, wherein in the deployed arrangement, the pair of inner panels are configured to define a planar boundary of confined space in which concrete is poured.
7. The collapsible concrete form of claim 1, further comprising a form coupling hole extending through one of the pair of outer flanges, the form coupling hole configured to receive a fastener for

coupling a respective outer flange of the pair of outer flanges to an adjacent form section of the pair of adjacent form sections.

8. A concrete forming system comprising: a collapsible form including: a pair of inner panels hingedly coupled to one another; a pair of outer flanges configured to abut an adjacent form section, a first one of the pair of outer flanges hingedly coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges hingedly coupled to a second one of the pair of inner panels; and a pair of waler bracket mounting holes configured to receive a waler bracket, a first one of the pair of waler bracket mounting holes extending through the first one of the pair of outer flanges and a second one of the pair of waler bracket mounting holes extending through the second one of the pair of outer flanges; and a pair of waler brackets configured to be inserted into the pair of waler bracket mounting holes, wherein the pair of inner panels and the pair of outer flanges are configured to move between a deployed arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew.

9. The concrete forming system of claim 8, further comprising: a locking brace protruding from one of the pair of inner panels; and a rigid member, wherein in the deployed arrangement the pair of waler brackets are configured to support the rigid member disposed between the pair of waler brackets and the locking brace, wherein when the rigid member is disposed between the pair of waler brackets and the locking brace, the pair of waler brackets and the locking brace abut the rigid member, preventing the pair of inner walls and the pair of outer flanges from moving from the deployed arrangement.

10. The concrete forming system of claim 9, wherein the locking brace includes: a pair of projections extending from one of the pair of inner panels; and a bearing panel extending between the pair of projections, the bearing panel disposed parallel to one of the pair of inner panels and configured to abut a rigid member disposed between the pair of waler brackets and the locking brace.

11. The concrete forming system of claim 9, wherein when the rigid member is disposed between the pair of waler brackets and the locking brace, the pair of outer flanges are configured to abut the rigid member.

12. The concrete forming system of claim 8, further comprising: a handle protruding from one of the pair of inner panels or a hinge coupling the pair of inner panels, the handle operable to move the pair of inner panels and the pair of outer flanges between the deployed arrangement and the collapsed arrangement.

13. The concrete forming system of claim 8, further comprising: a form coupling hole extending through one of the pair of outer flanges; a form section including: a form section panel; a frame coupled to the form section panel and disposed along a perimeter of the form section panel; and a form section coupling hole extending through the frame; and a fastener configured to be inserted through the form coupling hole and the form section coupling hole, coupling the collapsible form and the form section.

14. The concrete forming system of claim 13, wherein when the collapsible form and the form section are coupled, the pair of inner walls and the form section panel are collinear and collectively define planar boundary of a confined space in which concrete is poured.

15. The concrete forming system of claim 13, wherein the form section coupling hole comprises a pair of form section coupling holes, wherein the fastener is configured to be inserted through the form coupling hole and one of the pair of form section coupling holes, coupling the collapsible form and the form section, and wherein one of the pair of waler brackets is configured to be inserted through one of the pair of waler bracket mounting holes and an other of the pair of form section coupling holes.

16. A collapsible concrete form comprising: an inner portion including a pair of inner panels pivotably coupled to one another, the inner portion configured to move between a linear arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which

the pair of inner walls are askew; and a pair of outer flanges configured to abut a pair of adjacent form sections, such that the pair of outer flanges and the inner portion are disposed between the pair of adjacent form sections, a first one of the pair of outer flanges pivotably coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges pivotably coupled to a second one of the pair inner panels.

17. The collapsible concrete form of claim 16, further comprising: a locking brace protruding from one of the pair of inner panels; and a pair of waler bracket mounting holes configured to receive a pair of waler brackets, a first one of the pair of waler bracket mounting holes extending through the first one of the pair of outer flanges and a second one of the pair of waler bracket mounting holes extending through the second one of the pair of outer flanges, wherein in the linear arrangement, the pair of waler brackets are configured to support a rigid member disposed between the pair of waler brackets and the locking brace and the pair of waler brackets and the locking brace are configured to abut the rigid member, preventing the pair of inner panels from moving from the linear arrangement.

18. The collapsible concrete form of claim 17, wherein the locking brace includes: a pair of projections extending from one of the pair of inner panels; and a bearing panel extending between the pair of projections and disposed parallel to one or the pair of inner panels, the bearing panel configured to abut a rigid member disposed between the pair of waler brackets and the locking brace.

19. The collapsible concrete form of claim 17, further comprising: a handle protruding from one of the pair of inner panels or a hinge coupling the pair of inner panes, the handle operable to move the pair of inner panels between the linear arrangement and the collapsed arrangement when the rigid member is not disposed between the pair of waler brackets and the locking brace.

20. The collapsible concrete form of claim 17, wherein when the inner portion is in the linear arrangement, the pair of inner walls define a planar boundary of confined space in which concrete is poured.
